

05 · What to control for — and what NOT to (pathmc)

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The business decision behind the statistics. Every effect estimate in every other notebook depends on a choice made *before* any model runs: **which variables to control for** (put in the model / adjust for). Get it wrong and the number is biased no matter how fancy the estimator. The instinct “throw in every column we have” is actively dangerous. This is the notebook that lets you tell a skeptical CMO *exactly why* you controlled for what you did.

1.0.1 The four roles a variable can play — and the rule for each

To “**control for**” (equivalently *adjust for*, *condition on*) a variable W means to hold it fixed when comparing treated vs untreated — e.g. by including it in the regression. Whether that helps or hurts depends entirely on W ’s causal role between treatment T and outcome Y :

- **Confounder** — a common cause of *both* T and Y (e.g. customer loyalty drives both who gets emailed and how much they spend). It opens a spurious “backdoor” path $T \leftarrow W \rightarrow Y$. -> **Control for it**. Failing to is the classic omitted-variable bias.
- **Mediator** — a variable *on the causal path* $T \rightarrow W \rightarrow Y$ (the effect flows through it). -> **Do NOT control for it** — you’d subtract out part of the very effect you’re trying to measure.
- **Collider** — a common *effect* of T and Y ($T \rightarrow W \leftarrow Y$). Conditioning on a collider *opens* a path that was closed, **manufacturing** a correlation that isn’t causal. -> **Do NOT**.
- **M-bias collider** — a *pre-treatment* variable that is a collider between two hidden causes. -> **Do NOT** — proof that “just control for everything measured before treatment” is wrong.

The tool that decides this formally is the **backdoor criterion** on the **DAG** (the causal graph): a valid **adjustment set** is any set of variables that blocks every backdoor path from T to Y *without* including a mediator or a descendant of T . **pathmc** reads the graph and enumerates the valid sets for us.

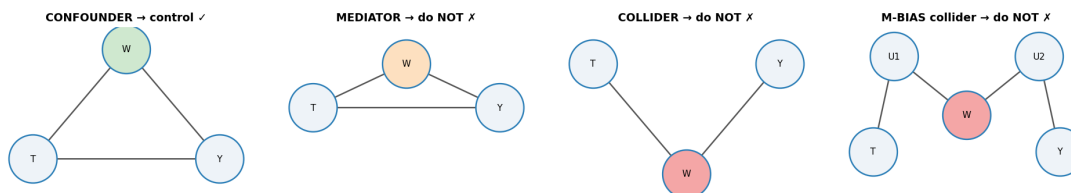
1.0.2 What this notebook does

- a **gallery** of the four structures with the correct action;
- pathmc** enumerates the admissible adjustment set and flags colliders straight from the DAG;
- we show the bias of each wrong choice **in euros**;
- we bound the residual risk from an *unobserved* confounder with a sensitivity contour and an **E-value** (the minimum confounder strength that could explain the effect away).

7-step contract, with **identification** (which variables make the effect recoverable) — not estimation — as the whole point.

1.1 1b · The four structures — a control-or-not gallery

Every “should I control for W?” question reduces to *what role W plays* between treatment T and outcome Y:



- **Confounder** (common cause of T and Y): **control** — it opens a backdoor path.
- **Mediator** (on the causal path T→W→Y): **do not** — you’d remove part of the effect you want.
- **Collider** (common effect of T and Y): **do not** — conditioning *opens* a spurious path.
- **M-bias collider** (pre-treatment, common effect of two latents that separately cause T and Y): **do not** — even a *pre-treatment* variable can be a trap. “Adjust for everything pre-treatment” is wrong.

1.2 2 · Simulate a ground truth

email → spend with a **true effect of €6**. Around it: **loyalty** is a **confounder** (drives both email and spend → must control); **responded** is a **collider** (caused by both email and spend → tempting but poisonous); **opened_email** is a post-treatment descendant.

TRUE effect of email on spend = €6.0

	email	loyalty	opened_email	responded	spend
0	1.0	1.101262	1.0	0.0	27.705363
1	0.0	0.338431	0.0	0.0	17.425501
2	0.0	-0.539972	0.0	0.0	9.602566
3	0.0	-1.260242	0.0	0.0	5.056443
4	1.0	-1.894621	0.0	0.0	7.175152

1.3 3 · Identify — let the DAG choose the adjustment set

The **backdoor criterion**: an admissible set Z blocks every backdoor path T→Y and contains no descendant of T. `pathmc` enumerates admissible sets and flags colliders straight from the graph — **no data needed**, because identification is about structure, not numbers.

```
Admissible adjustment set(s) for email -> spend: [{'loyalty'}]
Identifiable: True
```

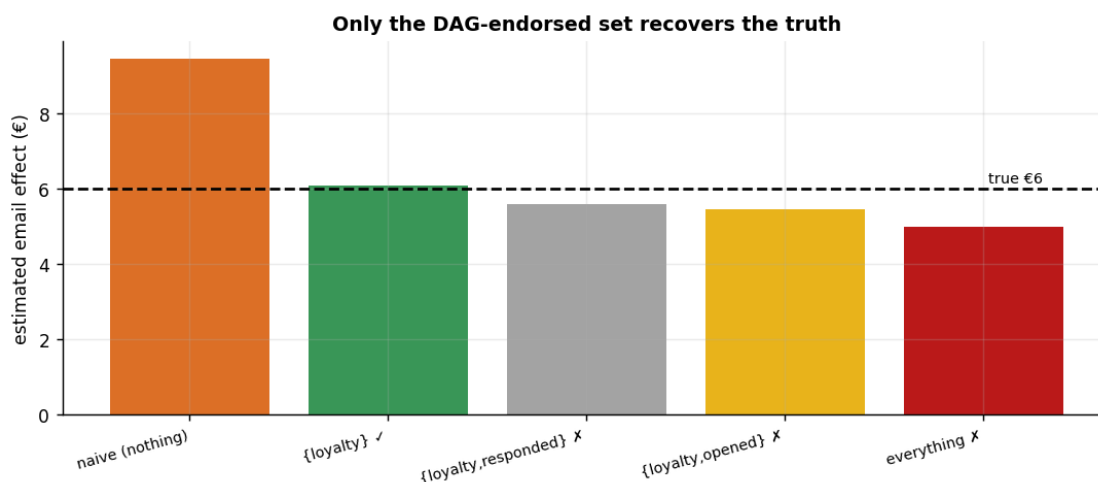
```
control for {'loyalty'}: OK - no collider opened
control for {'loyalty', 'responded'}: 'responded' is a collider between 'email'
and 'spend'. Conditioning on it may open a spurious path and introduce bias.
```

```
control for {'loyalty', 'opened_email'}: OK - no collider opened
control for {}: OK - no collider opened
```

1.4 4–5 · Estimate & Validate — every control choice, in €

We fit the email->spend effect under five control choices and compare to the true €6. Only the DAG-endorsed set (`{loyalty}`) recovers it; the others are biased, and **the sign of the bias is predictable from the structure** — omitting the confounder biases *up*, conditioning on a collider or post-treatment variable biases in the direction of the induced association.

naive (nothing)	€ 9.45	(error +3.45)
<code>{loyalty}</code> [ok]	€ 6.09	(error +0.09)
<code>{loyalty,responded}</code> [x]	€ 5.58	(error -0.42)
<code>{loyalty,opened}</code> [x]	€ 5.45	(error -0.55)
everything [x]	€ 4.97	(error -1.03)



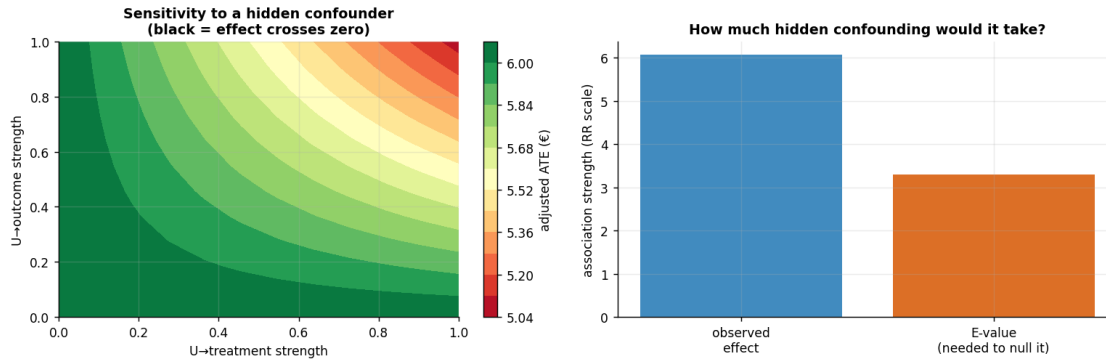
1.5 6 · Decide, in euros — with a robustness statement

The value here is a **defensible spec** we can explain to a skeptic: *here's why we controlled for loyalty, and here's why we did NOT control for opens/responses*. But identification is untestable, so we bound the residual risk: how strong would an *unobserved* confounder have to be to overturn the conclusion? `pathmc.sensitivity` gives the tipping surface; the **E-value** summarises it as a single interpretable number.

Observed adjusted ATE €6.07 (true €6.0) · E-value ~ 3.29

An unmeasured confounder would need ~3.3x association with BOTH email and spend to

explain the effect away. Ask the domain expert whether anything that strong is plausible.



1.6 6b · The stakes on real data (LaLonde) — good vs bad adjustment

This isn't an academic worry. On the famous **LaLonde** job-training data (fetched from a public URL — you provide nothing), the *right* control set recovers the known experimental effect (~\$1,800) while a careless one does not. It is the canonical demonstration that *what you adjust for* can flip the sign of your conclusion. Gated on `CMP_REAL=1`.

Real-data section skipped. Set `CMP_REAL=1` and re-run to fetch the LaLonde dataset.

1.7 7 · Caveats

- **The graph is an assumption.** `pathmc` gives the correct set *for the DAG you drew*. A missing or reversed edge changes the advice — the DAG must encode defensible domain knowledge.
- **Colliders hide in “obvious” controls.** “engaged”, “responded”, “opened”, “clicked” are almost always post-treatment; excluding them feels wrong to practitioners and is exactly right.
- **Front-door as a fallback.** If a T–Y confounder is genuinely unmeasured but a fully-mediating measured variable exists, the front-door adjustment can still identify the effect — a tool to keep in reserve.
- **Sensitivity != proof.** A high E-value means “robust to plausible hidden confounding,” not “unconfounded.” Pair it with the expert’s judgement about what could be missing.